PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Prefer Electric Corporation

Automation of Reports and Records

A Capstone Project

Presented to the Faculty of the

College of Information Technology Education

PHINMA University of Iloilo

In Partial Fulfillment

of the Requirements for the degree

Bachelor of Science in Information Technology

By:

Almonte, John LLoyd

Anaud, John Paul

July 2025

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled "Prefer Electric Corporation : Automation of Reports and Records" prepared and Submitted by John Lloyd Almonte, Jezza Joy Cabangal and John Paul Anaud here who are hereby recommended for corresponding final evaluation.

Ma.Abegail S.Labitag

Adviser

Approved by the Panel of Evaluators with a grade of PASSED

Dr. Arnold M. Fuentes, Ph. D, MPA, MMIT

Chairman

Andreas V. Armada

Member

Accepted in partial fulfillment of the requirements of the degree Bachelor of Science in Information Technology

Seth dA. Nono, MSCS

CITE, DEAN

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

ACKNOWLEDGEMENT

Researchers would like to extend our sincerest appreciation to those people who helped and guide us for the completion of this study.

Most importantly, God has given us the will to persevere, the concepts to keep working, and the hope to complete this study even when there are doubts and tests while working on this project.

To our teachers who mentored us without hesitation,Thank you so much for teaching us, guiding us, and understanding us for several trials. Without you all this project would not have been completed being grateful to have a teacher like you.

To our families, friends,and classmates, thank you so much for your generosity, sympathy,emotional support and financial support and pushing us through to keep working for this research study.

The proponents would want to express their gratitude to everyone who provided assistance and support throughout the process. None of this research would have been possible without your help. thank you enormously!

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

ABSTRACT

This research concentrates on report automation and record handling with a view of making data management processes more efficient, accurate, and well-organized. Manual processing of records and report creation within most institutions and workplaces usually results in delays, human mistakes, and challenges in following up information. Through the introduction of an automated system, the research looks at how technology can simplify tasks, ease workload, and provide easier access to required data.

The study entailed examining existing report and record-keeping practice, challenges identification, and conceiving an automated system that was user-specific. By testing and feedback, the automated system registered noteworthy improvements in processing time, uniformity, and general dependability of information stored. In general, this research emphasizes the advantages of digital transformation in record management and calls for the embrace of automated systems to facilitate improved decision-making and productivity in academic and professional environments.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

TABLE OF CONTENTS

Title page----------i

Approval Sheet----------ii

Acknowledgement----------iii

Abstract----------iv

Table of Contents----------v

List of Figures----------vi

List of Tables----------vii

CHAPTER I

Introduction----------

Background of the Study----------

Statement of the Problem----------

Objectives of the Study----------

Significance of the Study----------

Scope and Limitation----------

CHAPTER II

Literature Review----------

Overview----------

Traditional VS Modern System----------

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Technology Use----------

Case Studies----------

CHAPTER III

Methodology----------

Research Design----------

Model Use----------

Data flow Diagram----------

Entity Relationship Diagram----------

CHAPTER IV

Result and Discussion----------

System's Features----------

Implementation Challenges----------

User Feedback----------

Comparison to other System----------

CHAPTER V

Conclusion and Recommendation----------

Conclusion----------

Recommendation----------

Final Thoughts----------

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Appendices----------

Screenshots----------

Curriculum Vitae----------

Grammarian Certificate----------

LIST OF FIGURE

Figure 1 ----------SDLC Model

Figure 2 ---------- System Diagram Architecture

Figure 3 ----------Use Case Diagram

Figure 4 ---------- DFD Level 0

Figure 5 ---------- DFD Level 1

Figure 7 ----------ERD

LIST OF TABLES

Table 1 ---------- Justification of SDLC Method

Table 2 ----------Task Manager

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

CHAPTER 1

Introduction of the Study

The goal of this capstone project is to design and develop **Prefer Electric Corporation; Automated Reports and Records (A.R.R.)** to reduce human error and time delays caused by manual paperwork for reports and records. By integrating biometric systems with an online web application, the company can streamline administrative tasks and focus more on core operations, such as sales, service, and production of heavy electric machines. The platform will automate data entry, document generation, and record tracking through a centralized, secure system.

Automation of Reports and Records (A.R.R.) will feature a simple, web-based application enabling authorized personnel to generate, manage, and access multiple types of reports within just a few clicks. Integrated tools like automated report templates, real-time data synchronization, and electronic filing will eliminate the reliance on paper-based processes and prevent repetitive work. The system would issue notifications for pending or overdue reports, while providing departments with the ability to track document history, monitor compliance with company policies, and meet external regulatory requirements.

A secure database will ensure that sensitive corporate and client information is organized, protected, and easily retrievable.

By the end of this project, Automation of Reports and Records (A.R.R.) is expected to enhance **Prefer Electric Corporation's** operations by improving how data is collected, stored, and accessed across departments. Staff will be able to prepare reports more efficiently, track essential records in real time, and safeguard all critical information. Overall, A.R.R. aims to support smoother operations, reduce administrative bottlenecks, and strengthen decision-making processes within the company's sales, logistics, and after-sales service for heavy electric machinery.

Technical Background of the Study

Prefer Electric Corporation is a company specializing in the sales, distribution, and servicing of heavy electric machines. As a growing business in the industry, the company aims to deliver efficient services and reliable products to its clients. However, since its establishment, managing operations has been challenging due to the demands of modern technology. Many employees required training and adjustment to new systems, while

-the company faced the need to implement an efficient record-keeping and reporting process as soon as possible.

Using the old manual method of preparing reports, tracking sales, and monitoring employee records consumes a lot of time and effort. Issues often arise when employees and managers need to rely on physical time cards, handwritten logs, and manual filing systems to confirm attendance, commissions, or work performance. In today’s market, many businesses in Iloilo are already embracing modern technology for various purposes, such as inventory management, online payment systems, and digital record-keeping—areas where Prefer Electrical Corporation still lags behind.

To address these challenges, the company requires a platform that allows staff to create and send reports online while also monitoring employee records without relying on paper-based processes. Automation of Reports and Records (A.R.R.) was developed to provide this solution. This system enables the company to manage reports in a paperless environment and monitor employee performance with just a few clicks.

Overall, implementing this type of platform will make Prefer Electric Corporation’s operations more efficient, reducing manual work and improving productivity. By adopting automation in reports and records, the company can maintain high standards in employee performance monitoring while aligning its operations with modern technological advancements.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Implementing a **Prefer Electric Corporation; Automation of Reports and Records** is a crucial part of businesses to track their reports and records of their employees especially in Prefer Electric Corporation. Key technologies involved in such systems include:

1. **Database Management Systems (DBMS)**- Using modern and simple data management like MySql for storage and to be organized.

2. **Data Visualization Tools** - Usage of tools that generates visualized form of data to generate data reports.

3. **Server -** handles user requests, processes data, and ends back the appropriate response.

4. **Cloud computing -** Allows access, secured storage and share data anywhere.

5. **Automation Tools -** Streamline report generation and record-keeping capabilities.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Statement of the problem

The Prefer Electric Corporation main struggle is to monitor the employee's record,neglecting employee's report and behavioral performance manually:

1. Improve the behavior of Employee - It is absolutely imperative for the system to continuously work on enhancing employee behavior. This is critical as any gaps in employee conduct can result in false user satisfaction and feedback. It is essential that employees are proactive in adhering to the company's security guidelines to avoid any potential setbacks.

2. Data Inconsistency - The absence of standardized processes and uniform data structures among various sources, systems, or formats leads to disparities. Consequently, these disparities often cause discrepancies, inaccuracies, or information gaps within employee records, highlighting the critical need for cohesive data management practices to ensure data consistency and accuracy across all platforms.

3. Lack of user accessibility - The user's account might undergo changes based on monitoring by management. Therefore, it is possible that access to or editing of the user's account could occur if the management is overseeing the account for specific purposes or aspects, ensuring compliance and security measures are upheld.

4. Insufficient training for users – Therefore, the users are not sufficiently trained as the system provides comprehensive data on employee performance and satisfaction that they have recorded in the system, leading to a lack of sufficient skills and proficiency among the users utilizing the system.

5. Potential for system errors due to automation glitches – Although the system is not perfect, it is important to note that its functionality can be subject to variations leading to glitches. If these issues are not addressed promptly, it could potentially result in the mismanagement of employee reports.It is crucial for users to operate the system diligently and with precision.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Objective of the study

This study aims to design and develop Automation of Reports and Records (A.R.R.). A Modernized Approach to Automated Reports and Records System with streamlined upload reports in website applications overcome the delays of processing paper work loads and online monitoring. This capstone project aims to perform the following objectives:

1. **To create an efficient monitoring system** to reduce human errors and delays during working hours, to improve observation.

2. **To create a platform** for observation that can monitor behavior of each employee to find legal and acceptable evidence of not following the company rules.

3.**To design a system** that can easily be managed by designated staff of the office and user-friendly for the transportation employee.

4. **To develop an automated monitoring system** to avoid duplicating their reports on time and to monitor the date of starting the work. To increase the disciplinary action.

5.**To implement an online sending reports** to be monitored, reduce imbalance of attending and working diligently, secure the working station to avoid neglect-ions of work, and enhance worker development.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Significance of the study

The significance of this research extends beyond just the company itself, as it also contributes to the information technology sector by showcasing the emergence of a highly efficient and effective system for reporting and record-keeping. This not only sets a new standard for technological innovation in the industry but also highlights the potential for more streamlined and accurate data management practices moving forward.

This study revolves around the groundbreaking advancements made in the Prefer Electric Corporation, a cutting-edge system specifically designed to revolutionize traditional monitoring methods within the company. By streamlining and automating various tasks, this innovative system promises to significantly decrease the workload typically associated with monitoring processes.The key benefits of this study in developing Automation of Report and Records System include:

1. **Real-Time Access Monitoring -** Users can generate and access up-to-date reports and records instantly, enhancing decision-making and responsiveness.

2. **Increased Efficiency -** Automation can reduce manual data entry and processing time, allowing tasks to be completed faster and with less human effort.

3. **Cost Savings Over Time -** Although there is an initial development cost, automation reduces labor costs, paper usage and errors resulting in long-term savings.

4. **Scalability -** System can handle more data volume without decline performance ideal for department

5. **Environmental Sustainability** - Reduces the need for papers and printing, supporting green initiative and eco-friendly practices.

The beneficiaries of Prefer Electric Corporation ARR are the following:

1. Proponents - The automation will give the proponents benefits when writing a report and work performance. Instead, the traditional way of work will turn into automated and optimized work ethics while the other hand focuses on critical objectives in work.

2. Prefer Electric Corporation -An automated report and record management system will give the Prefer Electric Corporation team prompt access to accurate and updated reports for their work. This will enable sounder decision-making, support better tracking of employee performance, and create improved organization-wide transparency for employees. An automated report and record management system will also reduce the burden of manual work and the potential for human error when working with organizational data.

3. **Admin and Employees** – The administrative department and staff will benefit from a more orderly and effective method of tracking reports and records. The automation makes records and reporting easier for the administration by lessening manual effort, providing easy and secure access to information storage and retrieval. This enhancement enables staff to focus on more valuable duties, and arguably more important, enhances accuracy, productivity and efficiency in the organization.

Scope and Limitation of the study

This study focuses on an online system created to streamline the handling of employee documents and reports in the company. The system encompasses essential functions like attendance tracking, performance assessment, and automated report creation. Although it provides enhanced efficiency and precision, it also poses issues concerning data security, employee privacy, and reliance on the internet. System performance depends on a reliable connection and adequate upkeep to guarantee seamless access and safeguard against cyber threats. Consistent updates and oversight by the HR management team are crucial to ensure dependability and protect employee data.The features will be:

1. **Attendance Record/Tracking -** Monitoring users/employees through online tracking their presence,absences and punctuality.

2. **Overall Performance Evaluation -** Assessing user output based on the criteria while using a visualization tool like a pie chart,bar graph, and gantt chart to track users performance and behavior.

3. **Automated Report Generation -** Creating an organized detailed report of users quickly and accurately.

4. **Operational Efficiency -** Minimizing manual efforts and enhancing the efficiency of HR tasks. This system aims to assist HR managers and office personnel in managing records effectively, facilitating informed decision-making, and ensuring accurate and secure documentation.

Limitation of the study

While the study provides numerous benefits, it also presents various limitations:

1. Privacy and Monitoring Risks - involves monitoring staff behavior and activities. If implemented incorrectly or in the absence of clear policies, it may violate privacy rights or lower workers' trust levels.

2. Internet Dependence - As a web-based system, it requires a consistent internet connection and appropriate devices. Any loss of connectivity can make the system offline for temporary periods.

3. Cybersecurity Vulnerabilities - Lack of strong data protection means the system may be open to hacking, data loss,

or unauthorized access. Regular updates and safe protocols must be in place to keep sensitive information safe.

4. Maintenance Requirements - Maintenance of the system includes routine updating and technical support to ensure that the software works properly and remains secure. Delays in maintenance can result in problems with performance.

5. Risk of Misuse - The platform must not be used to produce or publish inappropriate content, such as sexual content, hate speech, or violent content. Adherence to organizational and legal conditions is necessary for proper system use.

Definition of Terms

1.Automation – Indicates the application of technology to execute tasks requiring little human involvement. This research emphasizes automation in the efficient generation and management of reports and records using computerized methods.

2.Reports – Organized documents that convey structured data or information regarding employee performance, attendance, and various work-related activities. These reports are automatically created by the system using the saved records.

3.Records – Electronic files or data entries that hold vital employee information including attendance, performance, and job specifics. These records are kept and organized within the database of the system.

4.Web-Based System – A platform that can be accessed via an internet browser, enabling users to manage and view reports and records at any time and from any location, provided there is a reliable internet connection.

5.Database – An organized set of data overseen by the system to safely store, arrange, and access employee reports and records.

6.User Interface (UI) – The graphical arrangement and design of the system that enables users to engage with its functionalities effortlessly and effectively.

7.Performance Assessment – The method of evaluating employee work effectiveness using data and reports produced by the system to aid in decision-making and track productivity

Chapter 2
Review of Related Literature

Overview of the Automation of Reports and Records

Maintaining accurate employee attendance and work records is essential in any organization. These records help management recognize diligent and reliable employees while ensuring fair performance evaluation. Automating report generation streamlines the process and provides real-time data for tracking attendance and productivity. With reliable records, companies can identify areas for improvement, reward high-performing staff, and make informed decisions that support overall organizational success.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Traditional vs. Modern Reports and Records

Traditional monitoring systems rely heavily on manual processes, where human operators are tasked with collecting and analyzing data. This manual approach can often lead to errors and inefficiencies, resulting in wasted time and effort.

On the other hand, modern monitoring systems take advantage of advanced technologies such as workflows, blockchain, and cloud computing to streamline and accelerate operations. By automating certain tasks and utilizing innovative tools, modern systems are able to operate more efficiently and effectively, reducing the likelihood of human error and optimizing overall performance. This shift towards using advanced technologies has revolutionized the way monitoring systems function, enabling organizations to achieve higher levels of accuracy and productivity in their monitoring processes.

Technologies

The modern reports and records used in the companies will include several key technologies to ensure efficiency and effectively:

1.Cloud Storage – Securely stores employee records online, accessible from any device, ideal for remote or traveling staff.

2.TIMS (Time Information Management System) – Tracks employee performance, attendance, and productivity, providing insights for HR and management.

3.File Servers – Centralized system for storing and sharing employee reports, enabling collaboration.

4.Databases – Structured storage for organizing and efficiently retrieving employee records.

5.Document Management Systems (DMS) – To manage document reports and track documents digitally, reducing paper usage and improving accessibility.

According to **Gayareth (2025)**, the Automated Document Digitization System: Handwritten Text to Excel aims to transform handwritten notes or logbook entries into structured digital spreadsheets. The research emphasizes that automation can greatly decrease the time and labor needed for handling paper records. By converting manual inputs into Excel files, organizations can preserve data precision, guarantee simpler access to information, and encourage environmentally friendly, paperless practices. This system further reduces the risk of data loss from physical document damage and enhances the efficiency of data retrieval and reporting procedures

Gayathri, B., Hariharan, S., & Nithya, S. (2025). *Automated document digitization system: Handwritten text to Excel.* Tech Innovations Publishing. ISBN 979-8-3315-2243-8

John Wiley & Sons (2007), stated that the Automated Defect Prevention: Best Practices in Software Management the significance of defining roles, choosing suitable technology, and creating clear interactions between individuals and systems

to enhance project management. The book highlights how automation can simplify repetitive tasks, coordinate project activities, monitor progress, and gather crucial project data for informed decision-making. Combining human roles with technological processes, the method minimizes mistakes, guarantees ongoing oversight, and offers practical insights for efficient management. This research highlights the significance of automation in enhancing workflow efficiency, which is directly applicable to creating systems for automated reporting and recordkeeping.

John Wiley & Sons. (2007). *Automated defect prevention: Best practices in software management.* John Wiley & Sons. ISBN 978-0-4701-6516-4

Shermis and Burstein (2003) stated in their book that Automated Essay Scoring: A Cross-Disciplinary Perspective, which offers an in-depth examination of automated essay scoring systems and their assessment across various fields, such as education, testing and measurement, cognitive science, computer science, and computational linguistics. The research underscores how automation provides uniform, effective, and impartial evaluation of written assignments, minimizing human bias and enhancing assessment precision. Integrating evaluation algorithms with tech tools, automated scoring systems enhance the grading process while ensuring high reliability. This study shows the capability of automation in handling extensive data, a concept that can be utilized for automating reports and documentation in business settings.

Shermis, M. D., & Burstein, J. C. (2003). *Automated essay scoring: A cross-disciplinary perspective.* Routledge. ISBN 978-1-4106-0686-0

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Case Studies

In various scenarios, the introduction and integration of automation tools at Prefer Electric Corporation have unquestionably proved to be a transformative factor for the organization. Through the utilization of automation-generated reports and systematic record-keeping, Prefer Electric Corporation has experienced notable enhancements in the efficiency of managing employee records and reports, which has consequently simplified the day-to-day operational processes.

The adoption of this advanced technology has facilitated the seamless transition to digitizing and consolidating employee data, establishing a centralized system that offers quick access and streamlined management solutions.Consequently, tasks that previously demanded a considerable amount of manual labor and time investments are now effortlessly completed, allowing employees the opportunity to dedicate their efforts towards more strategic and forward-thinking projects.

Chapter 3

Methodology

This project will utilize an array of qualitative methods to obtain real time data from users, specifically focusing on automation in reports and records. The project aims to obtain information from participants quickly utilizing automated tools and technology. By concentrating on automation in data collection, this research project will streamline data collection for accurate and efficient collection. This means researchers will have greater access to valuable information from users faster and more systematically, improving the quality and utility of the research findings.

Business Informatics also followed the System Development Life Cycle (SDLC) to ensure accuracy and continuous improvement of features during usability testing. Business Informatics utilized a variety of tools and technologies in areas of MySQL database management, user interface design, and backend programming. This process would present a thorough approach and ensure reliability and efficiency of an automation system that files, generates, stores, and retrieves company reports and records easily. This research study also considers secure data governance and system updates to assist in the performance of the system and the protection of sensitive information.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Research and Development Stage

The Research and Development (R&D) stage is a crucial part of the creation of the Automation of Reports and Records System(ARRS). The researchers will investigate what the current manual procedures for dealing with reports and employee records are and highlight existing concerns regarding common delays, people not collecting data correctly, and issues with organizing data. This study will aid the research team in learning more about the organization’s needs and how we can improve speed and reliability through automation. The aim is to build a system that collects and records efficiently while improving accuracy, availability, and efficiency.

Once all necessary data is collected, the design process will commence with the selection of appropriate tools and technologies based on the requirements. This may involve cloud storage for accessing (this may be in a database) the required information and software for automated reporting to produce accurate reports. The design will focus on creating an application that is easy to use so that the users can navigate and complete their tasks easily.

During this phase of the R&D process, prototypes will be developed and tested to ensure that features are working correctly and the system reaches its intended outcomes. Testing and feedback will continuously be a component of your R&D phase to detect mistakes and to make necessary improvements. The researchers will

adjust aspects based on user feedback in order to enhance performance and reliability.

System Development Life Cycle method

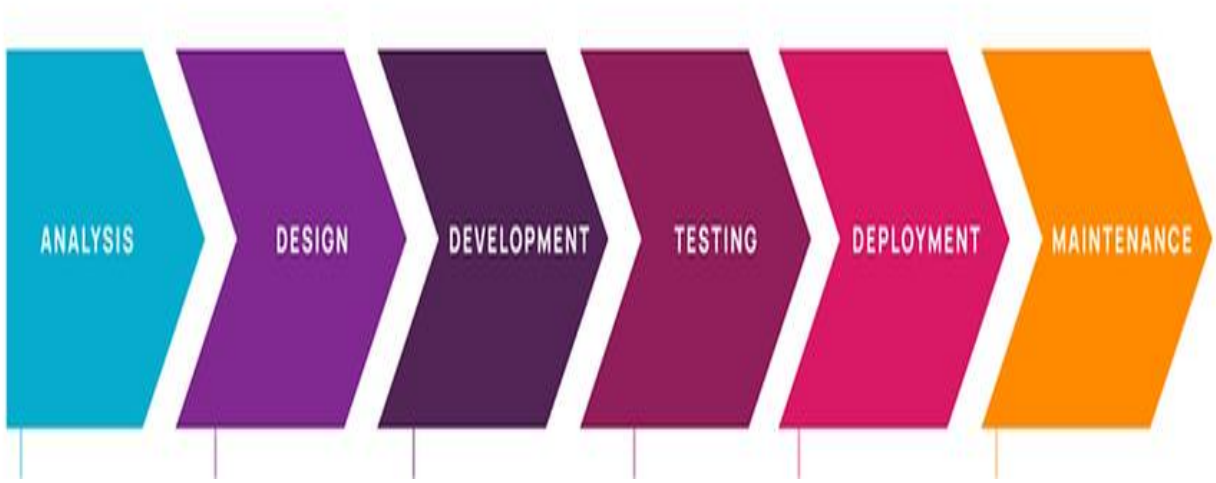


Figure 1. SDLC Model

The System Development Life Cycle was applied, featuring an iterative development process, with the project experiencing multiple enhancements driven by ongoing feedback. Every module of the automated reporting and record-keeping system was created in brief cycles or sprints, enabling quick testing and improvement. This method guaranteed that user requirements were frequently assessed and met, resulting in a more adaptable and user-focused system. It also reduced delays by detecting and addressing problems early in the development phase.

The following steps will make up the System Development Life Cycle Approach :

Planning: Begins with analyzing existing systems to identify missing requirements and productivity issues. This involves assessing workflows to uncover bottlenecks and inefficiencies.

By understanding these problems, organizations can create a clear strategy for improvement. The goal is to enhance overall efficiency and achieve better performance..

Analysis: is the systematic process of collecting, organizing, and interpreting data from reports, records, and other sources to support informed decision-making. It involves critical thinking and the ability to identify patterns and trends within the data. Through this process, analysts can reveal opportunities, detect risks, and suggest improvements. Ultimately, analysis aims to provide insights that guide strategic actions and support organizational goals.

Design: involves conceptualizing and developing the system's architecture, user interface, and database structure. It focuses on creating a seamless, user-friendly experience based on user needs, business goals, and technical limitations. The system architecture serves as a blueprint, defining how components interact and how information flows. A well-thought-out design ensures the system is both functional and intuitive

Implementation: is the phase where design concepts are transformed into functional software through coding, testing, and performance optimization. It focuses on developing a user-friendly and reliable system that meets organizational needs. This stage requires technical expertise to ensure the software is scalable and adheres to best practices. Special

attention is given to how the system supports workers' productivity, efficiency, and satisfaction in performing their tasks.

Testing: Through this technical monitoring process, the manager can identify areas where employees may need support, training, or process improvements. The data collected during the testing phase allows for evidence-based decision-making, helping to optimize workflows and eliminate inefficiencies. By leveraging these insights, the manager aims to foster a more productive and accountable work environment, ultimately enhancing the overall performance of the Prefer Electric Corporation workforce.

Maintenance : The maintenance stage of the Prefer Electric Corporation Automation System guarantees its ongoing reliability and effectiveness post-deployment. It includes resolving maintenance, modifying the system to meet new requirements, and enhancing functionalities based on user input. Routine preventive inspections will likewise be conducted to prevent possible problems. By engaging in these activities, the system will continue to be functional, secure, and in line with MITSCOOP’s operational requirements.

Data Collection Techniques

To obtain the information needed to design the Automation of Reports and Records System, a variety of data collection techniques will be employed. These methods aim to identify existing challenges in managing employee reports and records

whilst ensuring that the proposed system meets user needs, specifically for HR staff, administrative personnel, and other key stakeholders. Data will be collected using the following techniques::

1. **Interviews**

Interviews are conducted using human resource representatives, clerical staff, and employees who frequently manage reports and documentation. Using interviews permits the researchers to gain a full understanding of the challenges related to a manual documentation process and their expectations related to the suggested computerized system. By engaging with users directly, the team can identify specific features that may improve the efficiency of the system and the user-friendliness of the system. The interviews not only help develop rapport with employees but also help clarify in detail things that survey data or written documents may have not completely captured.

2. **Surveys**

Surveys are sent out to a wider group of employees for reported quantitative data on the current report and record management process. This method provides researchers with the ability to examine, even to a small degree, employee satisfaction levels, reaffirm patterns and speculate on employee readiness for the move to automation. Surveys can provide research teams greater context or detail in how employees go about the business process and help prioritize any system features based on user feedback.

Survey results are typically compiled and analyzed in order to gain insights that inform the design and features of the system.

3. **Document Evaluation**

The document review process involves looking at existing company documentation such as records, attendance logs, and report formats to understand the current documentation process. This process helps evaluate what types of information need to be automated or transmitted for usability, the type of formats and reporting standards should be used, and what needs to be improved. With the review of the documents, the researchers will be able to verify that the automated solution meets some of the reporting standards the organization already uses. Document review also builds in consistency of values in the data that was previously recorded manually and now recorded digitally.

4. **Observation**

The observation method permits researchers to see in real-time how employees create and process reports. By observing work performance, the team will identify repetitive tasks, bottlenecks, and potential sources of human error, which may be improved through automation. Actual observation provides an authentic perspective on work execution and helps create a product that meets the real needs of the user. Observations provide support for interview and survey findings.

Justification for Selecting the SDLC Methodology

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

SDLC was chosen over other Agile models, such Waterfall, for a number of important reasons:

ASPECTS	SDLC METHOD	TRADITIONAL APPROACH
Development Style	Uses an iterative and incremental development process, where the system is built and improved through multiple sprints or cycles.	Follows a linear and sequential process, completing one phase before starting the next.
Flexibility	High – easily adapts to changing company requirements	Low – changes are costly once phases are completed
User Involvement	Continuous feedback from Admin staffs and employees	Limited – mainly during initial requirements and testing
Delivery	Delivers working modules frequently for review and testing after each iteration.	Delivers the entire system only after the final phase is completed.
Risk Management	Risks are identified and resolved at every stage of development through	Risks are assessed at specific stages, often discovered late in the process.

	ongoing evaluation.	
Testing	Continuous testing ensures reliability and accuracy	Testing mostly at the end, issues detected late
Collaboration	High – Promotes strong collaboration between developers, users, and project managers.	Moderate – structured roles and limited communication
Adaptability to Change	Highly adaptable — supports rapid changes in features or user needs.	Poor – changes require revisiting completed phases
Efficiency	Reduces errors, rework, and speeds up development	Can be time-consuming and costly if errors arise late

System Architecture and Design

Overview

The Reports and Records Automated System is intended to efficiently oversee, review, and maintain reports on employees and operations within Prefer Electric Corporation. The system

allows for management and administrative persons to review the status of all reports, whether they are Submitted, Pending Approval, Reviewed, or Completed, so that the information is accurate, current, and available. In providing reliable processing, secure storage, and user-friendly access, the system is developed and designed using a multi-tiered architecture that separates three layers: the user interface layer, the business logic layer, and the data storage layer.

The system is designed to support multiple default accounts for administration only. The administration accounts are responsible for managing staff user access, system configurations, and the operation of reports and records notification. Allowing only the administration access to the default account provides security, accountability, and controlled access to sensitive organizational data.

The system integrates automated report generation, defined workflows, and performance maintenance, which leads to the tracking, reporting, and improvement of reporting working performance within the company. Employees' activities and reporting documents are recorded within the system, producing a quantifiable report for management to identify employees who may be high performers, as well as areas of reporting needing improvement. This approach provides an opportunity for work efficiencies, reliable accountability, and productivity to be maintained across the company, resulting in a more usable and effective system of management that is driven by performance.

System Architecture

The Report and Record Automation System is built using a three-tiered architecture, which will allow the system to be fully functional, allow the flow of data to be organized, and manage information in an organization securely. The use of a tiered architecture allows the system to scale, increases the ease of maintenance, and provides a clean separation between the user experience, application logic, and data storage. Furthermore, the tiers that incorporate automation allow reports and records to be processed in an efficient manner with accuracy and reliability.

Presentation Layer: The presentation layer allows interaction with the administrators of the system, who can monitor and update or delete reports and records when necessary. This layer will showcase a user-friendly and intuitive interface that allows the administrator to access and manage the data quickly and easily, maintaining order and responsiveness in the workflow.

Application Layer: The application layer is the core of the web-based application, processing, automating, and holding logic to create and manage reports and elements of record. The application layer ensures the input from users is verified, automated elements of the application work as intended, and overall application functions are seamless across the platform.

Data Layer: The data layer is responsible for securely storing all reports, records, and user data. The data layer ensures that

the data is protected and only accessible or modified through the application layer, ensuring confidentiality, integrity, and accuracy in stored content.

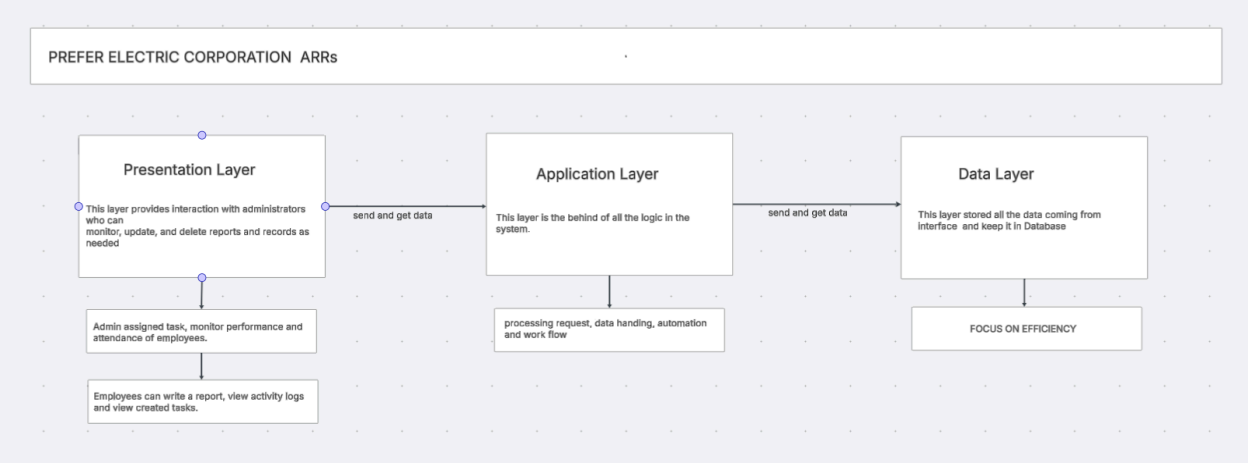


Figure 2 : System Architecture Design

System Components and Modules

The Reports and Records Automation System (RAR) aims to provide simple record management, consistent data tracking, and easy reporting to Prefer Electrical Engineering & Supply. The system consists of many modules that seamlessly integrate with other modules to ensure superior performance, reliability, and usability.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

1. **Records Management Module**

The Records Management Module is the centerpiece of the ARR, housing all employee and operational records.

Key Features:

- Creating, updating, editing, and deleting records of employees and departments.
- Ensuring accuracy and verifiable consistency to all data stored in the ARR.

2. **Reporting Generation Module**

The Reporting Generation Module takes the manual input out of report creation and formats the company reports automatically without the hassle of manual work.

Key Features:

- Preprogrammed daily, weekly, and monthly reports on employee performance and activity (attendance).
- Ability to convert data to a downloadable format (PDF, Excel, etc).
- Click and view dashboard for easy summary and quick overview.

3. **User Management Module**

The User Management Module operates as the control panel for users to access what is required only for authorized users to handle sensitive data.

Key Features:

- Admin access allows the admin to manage all records and reports as well as manage user roles and observe user activity.
- Employee access is specific to the employee's assigned reports and records.
- All user activities related to log in will be tracked on an activity page to maintain transparency and accountability.

4. **Analytics and Monitoring Module**

Delivers analysis of performance, attendance and productivity patterns across the organization.

Key Features:

- Produce analytics charts and performance summaries.
- Determine repeatable patterns or issues in performance.
- Inform decisions with data-driven analytics.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Use Case Diagram

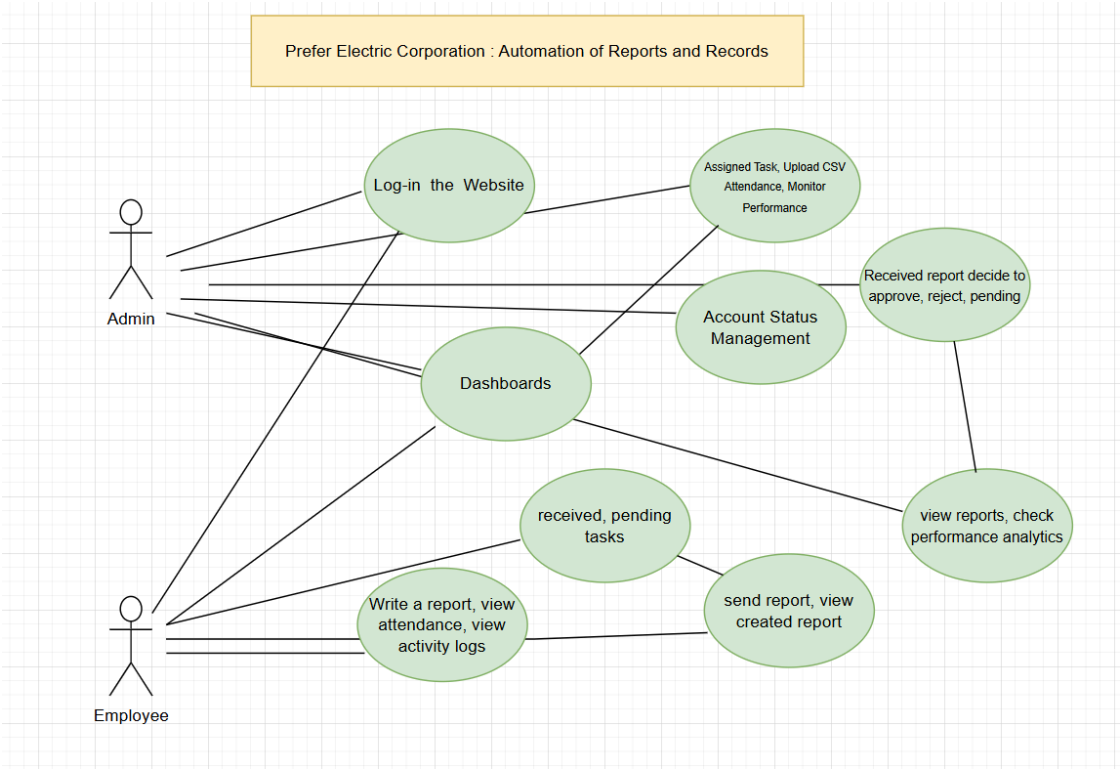


Figure 3 : Use Case Diagram

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Data Flow Diagram

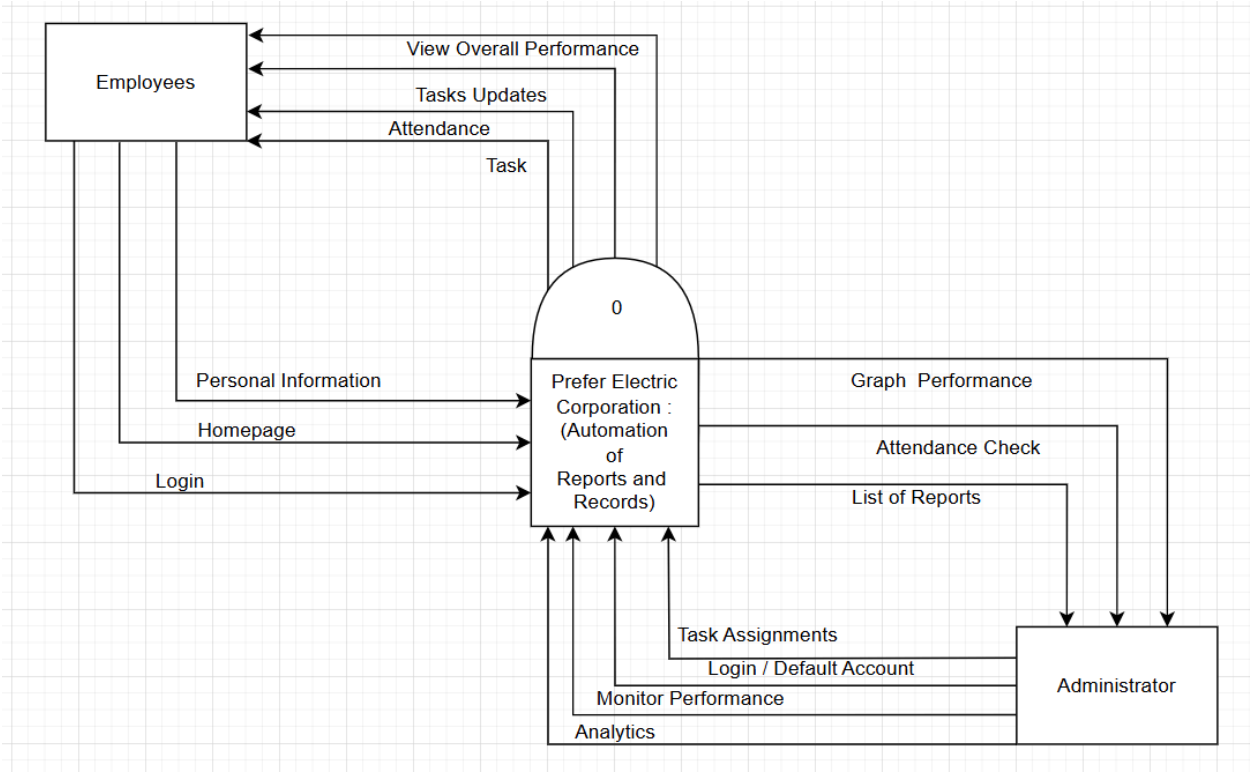
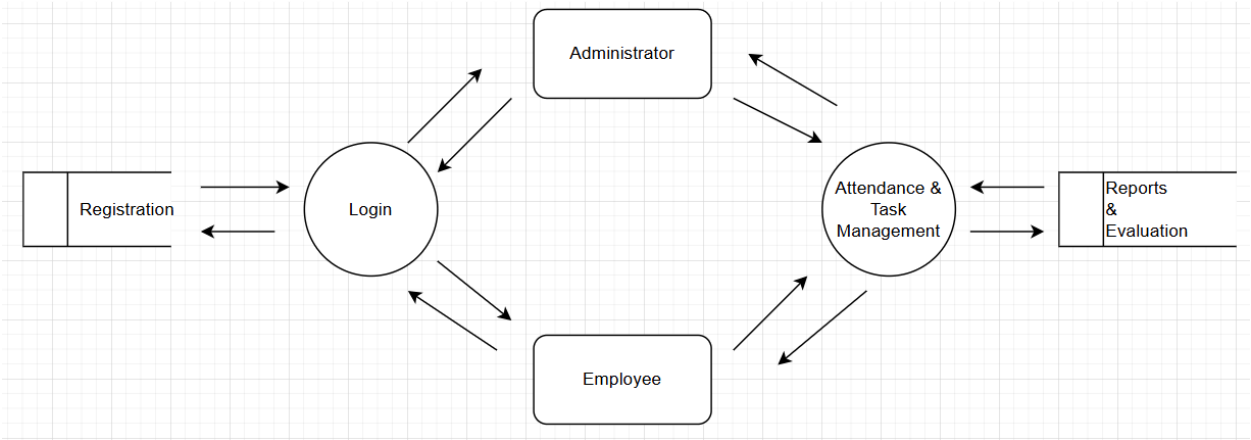


Figure 4 : Data Flow Diagram Level 0



PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Figure 5 : Data Flow Diagram Level 1

The Use Case Diagram and Data Flow Diagram (DFD) are essential elements in designing the Automation of Reports and Records System for Prefer Electric Corporation. The Use Case diagram illustrates the relationship between the users, namely Admins and Employees, and the primary actions of the system. The Use Case diagram visually depicts the interactions of each user with the system through actions such as logging in, report management, updating records, and performance tracking. This diagram also helps gather user requirements, outline system boundaries, and confirm that all actions are aligned with organizational objectives.

The Data Flow Diagram focuses on the movement of data throughout the system and shows where information comes from inputs, how data is processed, stored, and how it flows to outputs, typically as reports. This diagram allows developers to visualize the connection from one process to another, while ensuring data movement is performed correctly and efficiently. In addition, the DFD shows data flow communication between modules, allowing automated reports and records to be exchanged throughout the company seamlessly, and securely.

The Use Case Diagram and DFD provide a holistic view of how the system works, interacts between activities, and stores information. Collectively, they provide strength and clarity in design, reducing the likelihood of errors and supporting effective operations of the Automation of Reports and Records System to support Prefer Electric Corporation’s daily functions.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Entity Relationship Diagram :

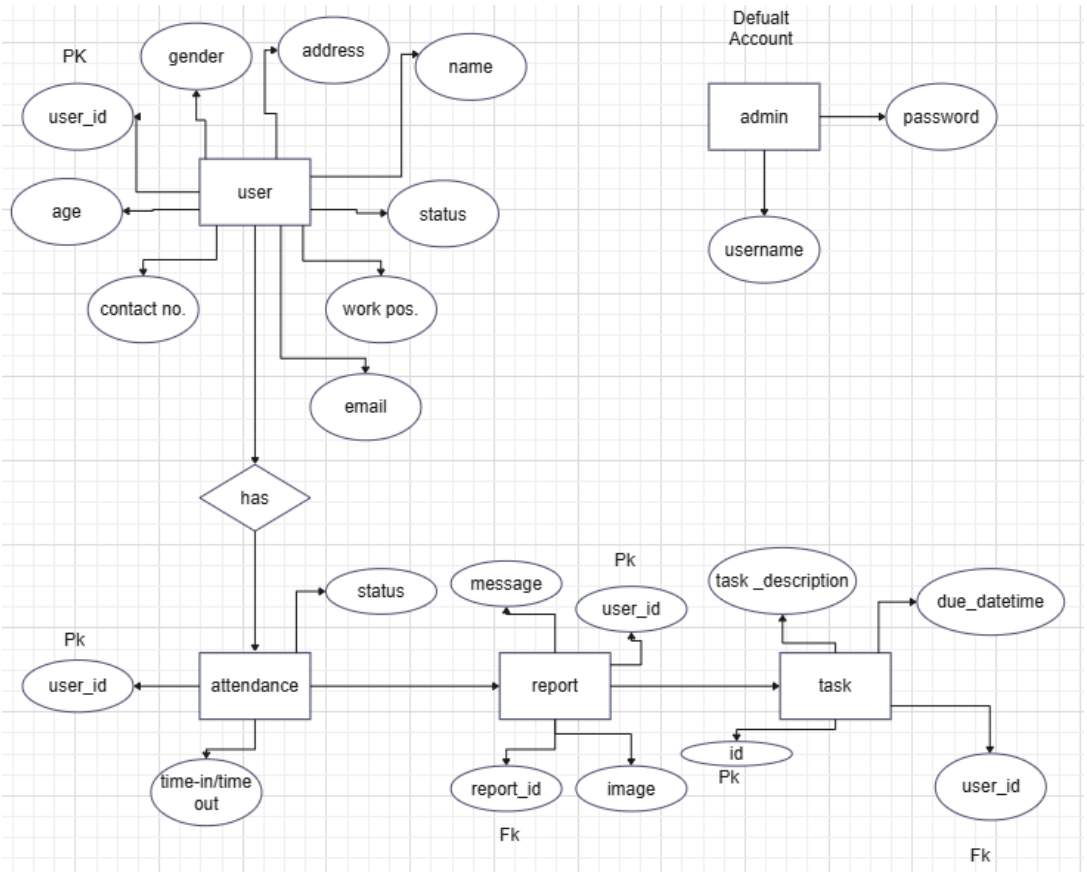


Figure 6 : Prefer Electric Corporation ARR ERD

Database Design

The design of the database for the Automation of Reports and Records System of Prefer Electric Corporation ensures the archival of data in an organized format, efficient retrieval, and the secure management of records. A relational model has

been used to maintain accuracy and reduce redundancies. The database key entities include Users(admin/employee accounts) Reports, Records and Performance Logs. Key tables include:

1.**Users** - A valuable registered entity inside the database like employees and their stored information. Fields: user_id, name, gender, address, age, status, contact, position, email, username, password

2.**Attendance** - It records daily attendance for employees, keeping track of what time the employee checked in with a status of present and absent. Fields: attendance_id, user_id, date, time, status

3.**Tasks** - Provides all details of tasks that were created by the admins for the employees that will show the deadline and the composed status of the task.Fields: task_id, user_id, task_description, due_datetime, status

4.**Reports** - store the report that was submitted by the employees on the task that was provided. Fields: report_id, user_id, message, image, status, task_description, report_date

5.**Account Evaluation** - Process the accounts performance by taking into thought, account attendance, usage of reports, and overall behavior to establish account status. Fields: id, name, attendance, reports, status

User Interface Design

The Automation of Reports and Records System for Prefer Electric Corporation was created with a systematic and user-centered method aimed at efficiency, accessibility, and organized

governance. The intent of the new system is to make the tracking of employee attendance, reports, and assignments easier for administrators and employees alike and to provide a useful means of documentation. The system was created as a web-based application, which guarantees interfaces that are responsive, clear, and navigable for daily office use.

The interface follows a general organization that is clean and appropriate for office conditions. The color scheme was simply created using a range of office colors that included soft tones that facilitate reading and ease of view for longer usage. The client agreed that the design and what the colors represent are tidy and professional, while considering that the functional components of the interface were clear and accessible. The colors are used sparingly to designate sections and to provide status on report submissions without overwhelming the greater user interface.

Administrators have access to attendance, assignment, report check and evaluation, and record management features, and the employees have visibility to attendance, can check to view assigned tasks, submit reports and see their history of activities. The modular design structure allows for extensibility in the future, enabling additional features to be added and integrated without interrupting the basic current functionality.

In conclusion, the Automation of Reports and Records System is a unique and web-based application developed with simple HTML and CSS for the front-end and Python Flask for the back-end. The application is intentionally simple and office-based, making it

an ideal interface for a professional and efficient environment focused on staff record-keeping and reporting. The client was pleased with the straightforward design, however efficient design supports client user understanding, effectiveness, and visual comfort in a clear space (line) for daily tasks and reports.

Security Considerations

1.**Password Hashing** - To safeguard user credentials, the system utilizes robust password hashing methods. User passwords are processed into secure hash values using algorithms such as bcrypt or Argon2 before stored in the database. This makes it almost impossible for an attacker to discover the original password even if the database is breached. This provides that sensitive login information remains private and secure from unauthorized access.

2.**Authentication Session** - Securely managing user sessions is an essential feature of the system to ensure that only authorized users can access sensitive data. Each time a user logs in, a one-time token is generated which will be used to authenticate every request to the sensitive data in order to prevent unauthorized access or session hijacking. The tokens will have an expiration time. Security of the user session is ensured by transmitting these tokens over HTTPS. Users are assured their session can remain secure while they engage with the system.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Technology Stack

1.**Frontend** – The system includes a frontend created in HTML and CSS, which resulted in a clean, intuitive and accessible interface for the user to view, add, and manage reports and records. HTML provides the structure for pages, forms, tables, and navigation menus, ensuring easy exploration of content and a logical organization of content. CSS is utilized for special styling to create a consistent color scheme, legible typography, and aesthetically pleasing layouts. The design led to a simple, user-friendly view for users to view, add, and manage reports. The system employs responsive design to ensure the system is functional and visually consistent on all resolutions and screen sizes for both desktop and mobile users.

2. **Backend** – The system's backend is built using Flask, a lightweight and flexible web framework written in Python. It manages core functionalities such as user authentication, task management, report generation, attendance tracking, and database connectivity. Flask's modular design promotes rapid development and seamless integration with the frontend. The backend consists of two main modules: one for user authentication and the other for the management of key backend functionalities - which leads to clear, organized, scalable, and maintainable code.

3.**Database** – MySQL is the core database management system for the application, responsible for effectively and reliably storing data in a structured way. It manages entities, including Users, Tasks, Reports, and Attendance records, ensuring proper

storage and retrieval of data. It securely stores user accounts and profiles, can track and assign task-based work in projects, can create reports and store generated records, and keeps attendance records accurately. MySQL ensures the integrity of the data while supporting fast response times for retrieving records, and the phpMyAdmin tool will facilitate web-based administration of databases.

4.**Additional Tools and Platforms**

- **GitHub -** is used as a version control system, allowing for the effective tracking and organization of code changes, as well as collaboration on different project versions..

- **Visual Studio Code (VS Code)**- which is primarily used for development purposes.

Data Gathering techniques

Survey Questionnaire for System Assessment

The survey intends to assess the capability and ease-of-use for the Prefer Electric Corporation- Automation of Reports and Records System. The survey is endeavoring to obtain user feedback regarding the system's user capacity, efficiency, and operations of the reporting and records system. Feedback from this survey will help pinpoint strengths and areas for improvement, ensuring the system continues to meet the needs of programs and the organization for reporting processes that are accurate, productive, and highly dependable

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Demographic Information

Name: _____
Age: _____
Position/Department: _____

1. What is your role in the company?
   - ☐ Employee/User
   - ☐ Administrative Staff
   - ☐ Department Head
   - ☐ IT Personnel
   - ☐ Other: _____

2. How often do you use the reporting or record management system?
   - ☐ Daily
   - ☐ Weekly
   - ☐ Monthly
   - ☐ Rarely

3. How familiar are you with the use of automated systems for reports and records management?
   - ☐ Very familiar
   - ☐ Somewhat familiar
   - ☐ Heard about it but not familiar
   - ☐ Not familiar at all

4. How important do you think it is to systematically organize and monitor company reports and records?
   - ☐ Very important
   - ☐ Important
   - ☐ Neutral
   - ☐ Slightly important
   - ☐ Not important

5. Do you believe that using an automated system can improve report generation and record accuracy?
   - ☐ Strongly agree

- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

6. How would you rate the current process of handling reports and records in the company (manual or existing system)?
   - ☐ Very efficient
   - ☐ Efficient
   - ☐ Neutral
   - ☐ Inefficient
   - ☐ Very inefficient

7. In your opinion, how necessary is it to replace the manual record-keeping process with an automated system?
   - ☐ Very necessary
   - ☐ Somewhat necessary
   - ☐ Neutral
   - ☐ Not necessary

8. Do you think an automated system can help reduce human error in managing reports and records?
   - ☐ Strongly agree
   - ☐ Agree
   - ☐ Neutral
   - ☐ Disagree
   - ☐ Strongly disagree

9. How do you usually report issues or concerns related to reports or records?
   - ☐ Inform immediate supervisor or admin staff
   - ☐ Report to IT department
   - ☐ Submit through a written form
   - ☐ Do not report
   - ☐ Other: _____

10. How long does it usually take before reported issues are addressed or resolved?
    - ☐ Within the day

- ☐ 1-3 days
- ☐ More than 3 days
- ☐ Not sure

11. What challenges do you commonly experience in managing or accessing company records?
    - ☐ Delayed reporting process
    - ☐ Missing or misplaced files
    - ☐ Data inconsistency or duplication
    - ☐ Limited system access
    - ☐ Lack of technical support
    - ☐ Other: _____

12. How important is it to have security measures that protect company reports and confidential data?
    - ☐ Very important
    - ☐ Important
    - ☐ Slightly important
    - ☐ Not important

13. Do you believe that an automated system can improve accountability among employees and admin staff?
    - ☐ Strongly agree
    - ☐ Agree
    - ☐ Neutral
    - ☐ Disagree
    - ☐ Strongly disagree

14. How confident would you feel if all company reports and records were stored and secured in a digital system?
    - ☐ Very confident
    - ☐ Confident
    - ☐ Neutral
    - ☐ Unconfident
    - ☐ Very unconfident

15. How likely do you think the Automation of Reports and Records System will improve efficiency and organization in company operations?

- ☐ Very likely
- ☐ Likely
- ☐ Unlikely
- ☐ Very unlikely

16. How helpful would it be if the system could automatically generate reports for audits, employee performance, or daily summaries?
    - ☐ Very helpful
    - ☐ Helpful
    - ☐ Slightly helpful
    - ☐ Not helpful

17. Do you think proper training should be provided to all employees and staff on how to use the system?
    - ☐ Strongly agree
    - ☐ Agree
    - ☐ Disagree
    - ☐ Strongly disagree

18. How beneficial is it to have automated notifications or reminders for pending reports or record submissions?
    - ☐ Very beneficial
    - ☐ Beneficial
    - ☐ Slightly beneficial
    - ☐ Not beneficial

19. Do you believe that automation can promote transparency and accuracy in company documentation?
    - ☐ Strongly agree
    - ☐ Agree
    - ☐ Neutral
    - ☐ Disagree
    - ☐ Strongly disagree

20. Overall, how important do you think implementing the Automation of Reports and Records System is for improving company operations?
    - ☐ Very important

- ☐ Important
- ☐ Neutral
- ☐ Slightly not important
- ☐ Not important

Sampling Methods for Respondents

This research applied the simple random sampling method to ensure fairness and impartiality of the participants. Several individuals who often use the report and records management systems were picked to participate and were asked to respond to the survey questionnaire. These individuals were picked from a population of actual users to ensure that the data collected reflects actual experiences and knowledge of the current processes.

This method of sampling was applied to give every potential participant an equal opportunity of participating in the research, thereby reducing bias and ensuring the reliability of the data. Furthermore, the focus on individuals who often manage reports and records ensures that the feedback collected is relevant and significant to the improvement of the automated system.

Although the number of respondents is limited, this method allows for more in-depth analysis of user perceptions, especially in relation to the efficiency, precision, and convenience of automated reports and records. The research aims to provide accurate and dependable findings that could help in improving and successfully implementing the proposed system.

Data Analysis Technique

The information gathered from the survey questionnaire will be assessed using descriptive and quantitative statistical methods in order to evaluate the effectiveness and level of user satisfaction with the Automation of Reports and Records System of Prefer Electric Corporation. The answers will be sorted, categorized, and coded, to obtain summary information on the usability and effectiveness as well as the overall performance of the system. The use of frequency and percentage distribution will allow for the summarization of respondent demographics and observe trends in their opinions about the system.

The feedback will be translated into numbers using a Likert scale, which will allow us to assess the extent to which respondents agreed or disagreed. Each response option from their choice will carry a weight value, therefore generating a weighted average. This will allow us to assess their overall satisfaction and the effectiveness of the system for automating reports and records for the company.

The results from the data analysis will be presented in table and graph format to provide a clear visual representation of the results. Conclusions and recommendations based on the study will be made to identify strengths of the system, opportunities for improvement, and user satisfaction. The evaluation will be used to inform design and functionality improvements to the system, ensuring that it helps Prefer Electric Corporation achieve more efficient, accurate, and reliable record keeping.

Ethical Consideration

The design and implementation of Automation of Reports and Records System for Prefer Electric Corporation underscores the

ethical and proper management of user information and company confidential information. All information that is obtained, processed, and stored in the Automation of Reports and Records System is treated with the utmost confidentiality to assure that private records cannot be accessed by anyone that is not authorized. The Automation of Reports and Records System is designed to protect employee information and company information from inappropriate use, alteration, or disclosures while ensuring integrity and accountability during the process.

According to the Data Privacy Act of 2012 (Republic Act No. 10173), this project serves as a commitment to manage personal data and organizational data safely. The statute also mandates organizations to adopt appropriate and effective measures to ensure the protection of data including the confidentiality, integrity, and availability of the information. The Automated Reports and Records System utilizes authentication mechanisms, access limitations, and encryption protocols to protect the user information from unauthorized access or compromise. Also, users become informed as to the extent and basis for data collection, thereby encouraging transparency and consent for all activities of information management.

This system, while implementing legal and ethical standards builds on Prefer Electric Corporation's commitment to safe, transparent, and accountable digital transformation.

Chapter 4

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Result and Interpretation

The implementation of the automated reporting and records system significantly improved the efficiency and accuracy of data management practices within the organization. Compared to the previous manual system, the new system improved report generation speed, decreased human error, and streamlined record access. In user acceptance testing, some of users demonstrated improved usability of the application, and reported a reduction in time spent on documentation tasks.

Functionality testing showed the application could automatically generate reports from real-time input instead of manually, securely store records in a centralized database, and allow for usability dashboards to display data. A core function such as filtering, date-range selection, and automatic archiving processes were successfully conducted with minimal delays. The addition of user roles and access control ensured data integrity and confidentiality.

The automation effort addressed numerous concerns identified in the planning phase - most notably delays in generating reports, and inconsistencies in data entry. Once the automation took hold, management reported quicker decision-making resulting from the ability to access up-to-date and reliable reports. There was greater productivity among staff since automated solutions reduced redundant tasks and allowed employees to focus on more value-added tasks.

Overall, findings from the project's implementation validate the overarching purpose of the project: to improve and modernize reporting, and documentation processes. The automation not only improved processes but also coordinated with the organization's longer-term goals related to digitization and sustainability. The findings highlight the importance of investing in digital solutions to address shifting organizational needs.

Task Manager

Gantt Chart / Project Timetable – Prefer Electric Corporation : Automation of Reports and Records (ARR)

WEEK	TASK	UI/UX	BACKEND	QA	DOCUMENTER
Week 1-2	Project Planning & Requirement Gathering	_____	_____	_____	Gather system requirements, create proposal
Week 3-4	UI/UX Design & Wireframing	Create layout using Html, CSS	_____	_____	Document wireframes and design drafts
Week 5-8	Frontend Development	Develop Dashboard, Login, Registration pages	_____	_____	Update documentation of progress
Week 9-12	Backend Development		Develop Flask		Record database

		_____	server, MySQL database,	_____	schema and backend setup
Week 13-16	**S**ystem Integration	Connect frontend to backend	Test data flow, ensure connectivity	_____	Document Integration Process
Week 17-20	Testing & Debugging	Fix UI/UX issues	Debug backend for logic and data	Conduct functionality test	Record test cases and results
Week 21-22	System Evaluation	Apply feedback and UI improvements	Refine backend performance	Retest after adjustments	Summarize evaluation results
Week 23-24	Documentation Finalization	_____	_____	_____	Complete final documentation and report
Week 25-26	Presentation & Submission	Prepare frontend demo	Demonstrate backend features	Assist during system demo	Compile and finalize deliverable

Table 2 : Task Manager

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Presentation of Results

The Management, administrative staff, and selected employees of Prefer Electric Corporation were provided a formal demonstration of the Reports and Records Automation System. The purpose of the demonstration was to show how the system automates and makes more efficient reporting and record-keeping tasks that were originally difficult, labor-intensive, and subject to human errors. The project team demonstrated each module, including the user authentication, report creation, record management, and activity logging components. In the demonstration, we showed how the system persists data and how it can - quickly and accurately - retrieve and organize it for the user. With this system in place, both employees and managers had enhanced tools which allowed them to speed up their work, and approach it with greater accuracy.In the system demonstration, the attendees had the chance to play with the interface, trying out key features they would explore further as users later. The interface had a dashboard page that listed reports and employee files, which served as a dashboard and/or summary of recent reports and records in the system. With the report generation feature, the data was automatically condensed and organized for review once

the user prompted those features - again making their work easier and more efficient. Administrative users had explicit instruction about account management and their access to duplicate activity log features for tracking activities, and how they could generate reports related to employee performance. Participant feedback reported the interface was easy to engage with, automated tasks almost completely eliminated repetition in their work. Along with the system speed, the security features which requested a login and verification at several points were noted as an upgrade from the manual processes.The survey received after the presentation showed very high user satisfaction for the Automation of Reports and Records System. Users agreed that it improved accuracy, access, and organization. The system mitigated the delays with retrieving data, improved workflow, and received high ratings for usability and security; meet operational needs while supporting Prefer Electric's digital transformation.

Impact of Prefer Electric Corporation : Automation of Reports and Records

By transitioning from paper-based to data-driven systems, the automation of reports and records has had a dramatic impact on the IT industry. Prefer Electric Corporation, who was previously

using paper to report information for much of their history, has taken a significant step toward modernizing their systems through automation. The result of implementing reporting was faster report production, increased data accuracy, and enhanced information accessibility, all of which improved internal processes and visually demonstrated the extent to which IT capabilities can transform business practices. In order to implement reporting, it became clear there was a greater need for enterprise-level software and data solutions that allow firms, like Prefer Electric, to appropriately adapt this broader IT trend into the workflow of their organization. Prefer Electric designed and implemented an organized database and cohesive reporting system for permanently managing different operational data, aligning with enterprise IT trends focused on designing solutions with scalability and efficiency for monitoring operational data in near real time. In this process of automation, data security and data compliance developed into paramount issues as information became automatically transferred, and was therefore necessary to develop a database and reporting system with formally defined access protocols. Data security protocols enforced the level of confidentiality of information and security protocols were typically chosen due to the lack of adherence to data privacy legislation in the past.

This expanded our responsibilities towards ethical obligations in the IT industry. In conclusion, Prefer Electric's exploration into the adoption of automation demonstrates the important role that IT played and will increasingly play in fostering efficiency, mitigated error, and digital transformation in organizations. Timely, the industry's general acceptance of automation identifies it as an element of widespread significance in the near future.

Contribution to Technological Advancements

By developing a Reports and Records Automation System for Prefer Electric Corporation, we made a large step toward delivering automation and moving away from manual paper reporting. The web-based system is a step toward operational efficiency, reducing processes and redundant human data error. The system also grants employees and administrators ease-of-use and access because it does not require specified advanced technology skills. Reporting performance can also be monitored in real-time and tracked as well as visible user engagement, so employees and administrators know who has viewed and/or updated a User Report.

Report generation and user assessment (comments) tools generate applicable report data for strategic management decisions. From an IT perspective, this case demonstrates how this automation tool illustrates the organization's transition from traditional office practices to a modern digital systems and infrastructure. Importantly, the case demonstrates how web-based technology can be used for accessibility, with the added benefit of better data validity, accuracy, security, and general administrative demeanor.

The takeaways and significance of this project are that small to medium-sized businesses can adopt and use IT solutions for a digital transformation with minimal, affordable costs---serving as a stepping stone for continued innovation and ongoing organizational efficiency---while fostering smarter, more connected essentially, Technology inter-, smart business processes and willingness to adapt.

Project Structured

```
 Prefer_Electric_Corp-Caps/
\---templates
    |  admin_add_task.html
    | admin_check_attendance.html
```

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

```
|  admin_dashboard.html

| admin_list_report.html

|  admin_record.html

| all_reports.html

| attendance.html

| check.html

| created_report.html

| home.html

| landing.html

| list_report.html

| logs.html

| monitor.html

|  profile.html

| register.html

| report.html

|test.attendance.html

|view_report.html

/--backend-

| main.py
```

Chapter 5

Summary,Conclusion and Recommendations

Summary

The purpose of the Automation of Reports and Records System developed by Prefer Electric Corporation is to modernize traditional reporting and records. The organization had traditionally operated using a manual process to develop reports, which resulted in delays, inaccuracies, and challenges retrieving information. The automated reporting system takes the reporting process and the records process, including the retrieval of information, and integrates these processes into a single electronic repository. This aids in improving accuracy of records, as well as the efficiency of the team since data sharing can be conducted in a timelier manner with an electronic report. The automated report and records repository have ease-of-use functionality, immediate metrics or key performance indicators, and sound data security authenticated and data protected in compliance with authorized standards. The use of the automated reporting system provides timely and accurate information when the information is most removed from the decision-to-be-made. The implementation of an automated reporting system is reflective of a journey of digitization characterized by an organization focusing on enhanced

efficiency, transparency, and accuracy on the organization's report.

Conclusion

In summary, the “PREFER ELECTRICAL ENGINEERING & SUPPLY: Automation of Reports and Records” system has demonstrated its effectiveness in enhancing the company's documentation and reporting procedures. The system effectively decreased manual work, lessened mistakes, and improved data precision and availability. Its straightforward design and automated functionalities offer users a practical and effective method for handling reports and records. In general, the project reached its objective of establishing a dependable and easy-to-use platform that enhances the company’s operational effectiveness.

Recommendations

Future enhancements should concentrate on boosting system performance and integrating advanced features like data analytics, cloud backup, and real-time alerts. Consistent updates and maintenance ought to be carried out to guarantee the

system stays secure and adapts to the company's changing requirements. Offering brief user training or orientation would additionally assist employees in fully leveraging the advantages of the system. Through ongoing enhancements and upkeep of the platform, the company can secure lasting success and viability of the automated reporting and record management system.

Reference (APA format)

1.Gayathri, B., Hariharan, S., & Nithya, S. (2025). Automated document digitization system: Handwritten text to Excel. Tech Innovations Publishing. ISBN 979-8-3315-2243-8

2.John Wiley & Sons. (2007). Automated defect prevention: Best practices in software management. John Wiley & Sons. ISBN 978-0-4701-6516-4

3.Shermis, M. D., & Burstein, J. C. (2003). Automated essay scoring: A cross-disciplinary perspective. Routledge. ISBN 978-1-4106-0686-0

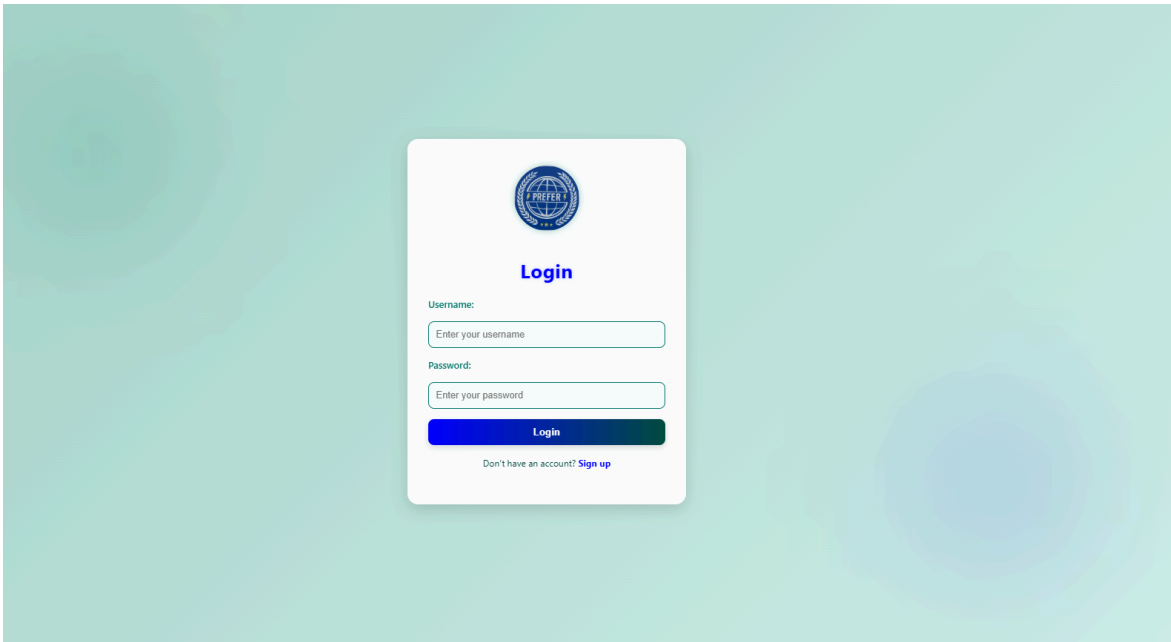
Appendices :

Appendix A : System Interface Screenshot

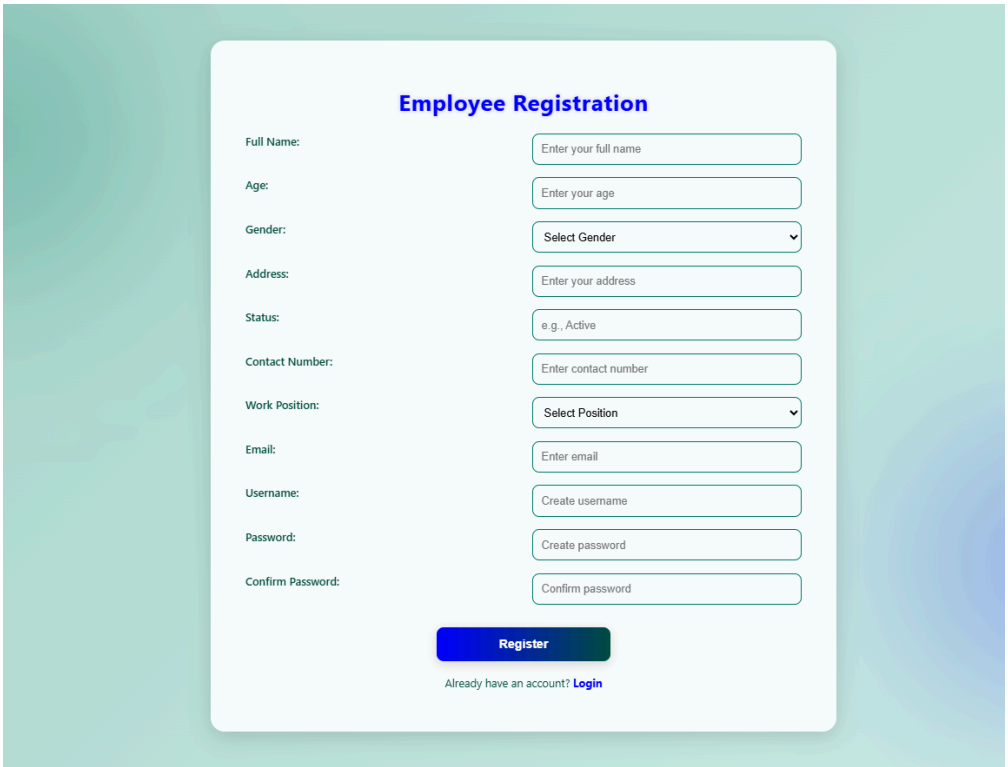
- **Login Page**

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION



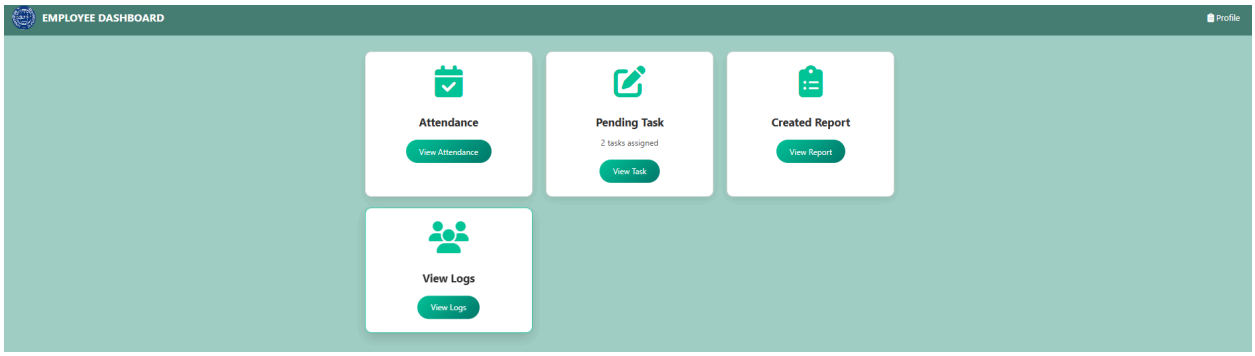
- Registration



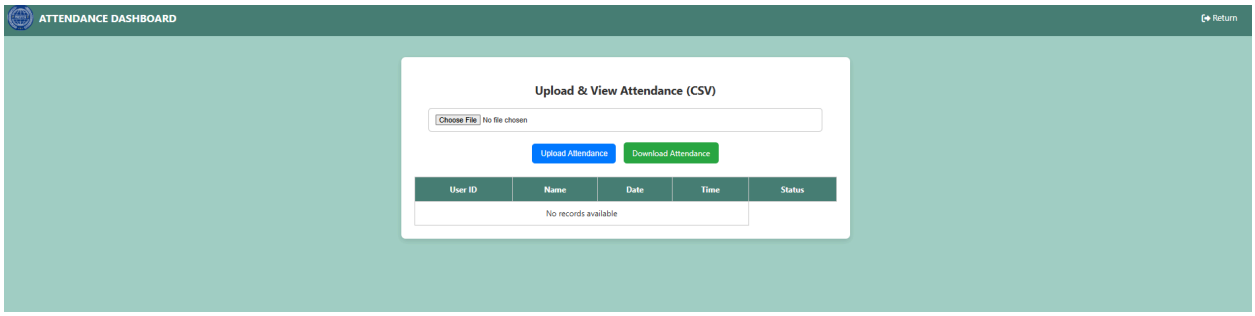
PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

- **Dashboards (admin/employee)**



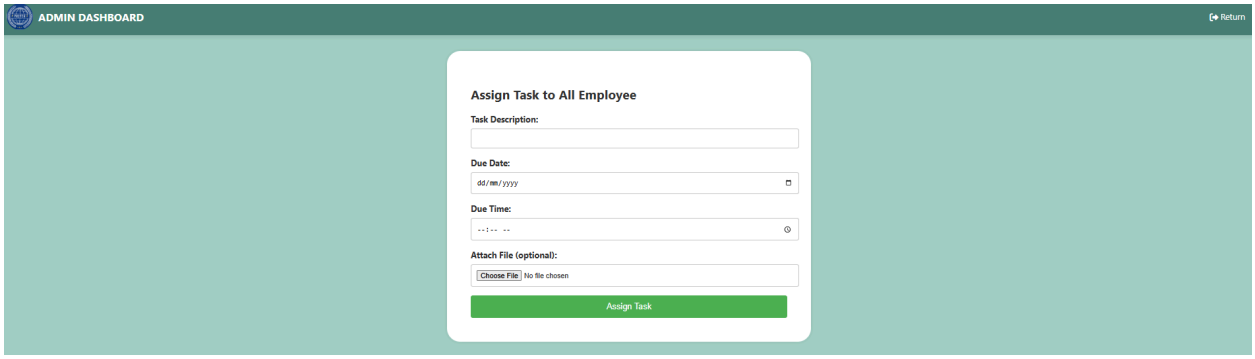
- **Attendance Records (ADMIN)**



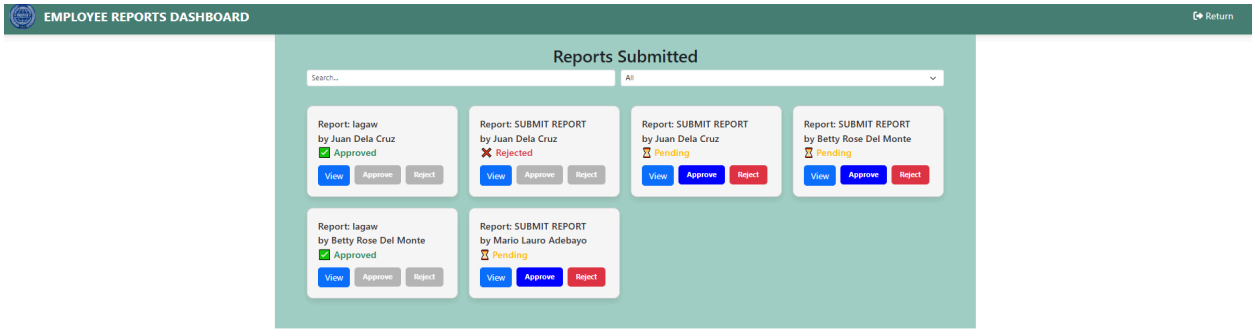
- **Add Task**

PHINMA UNIVERSITY OF ILOILO

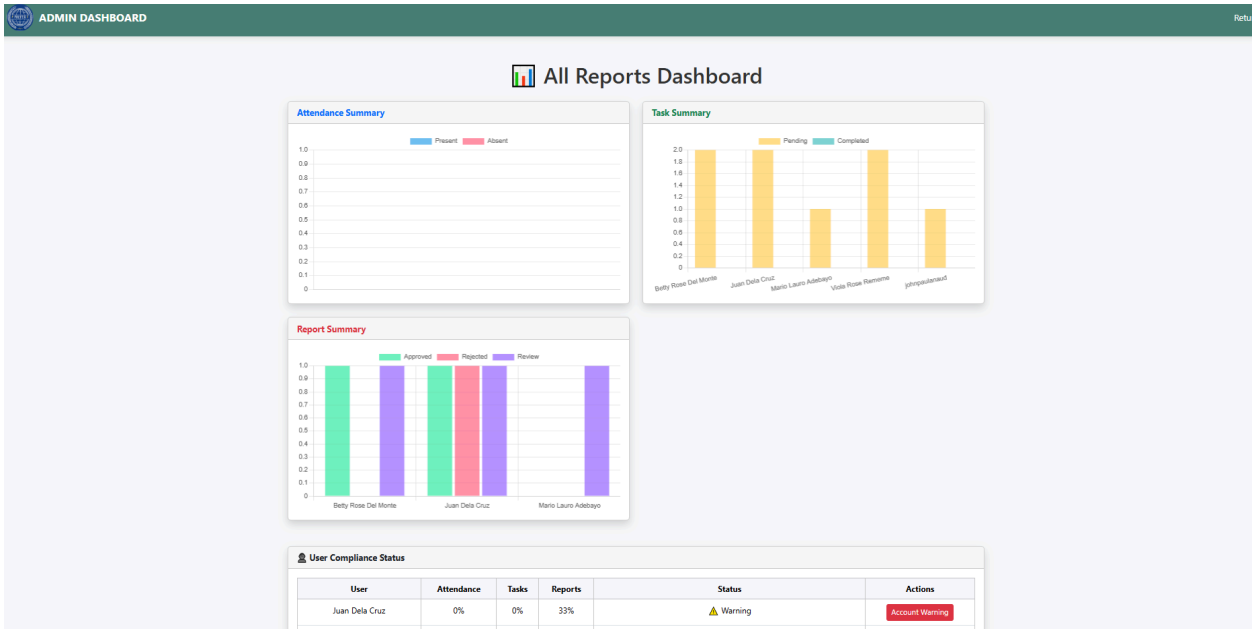
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION



- **View Reports**



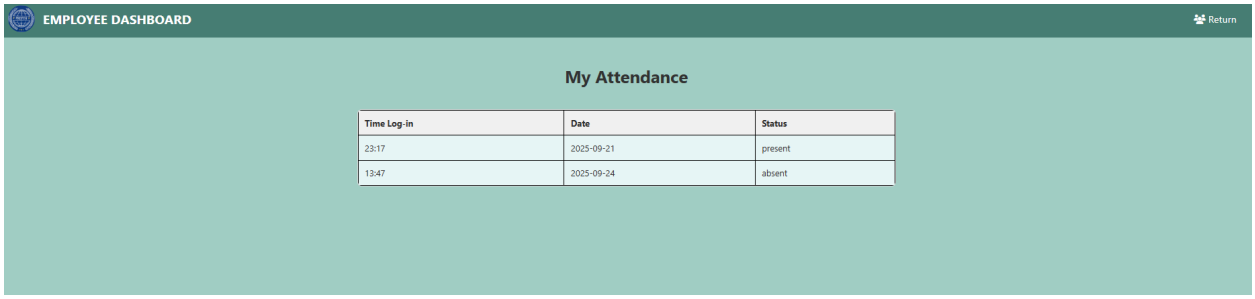
- **Evaluation (All Reports & Records)**



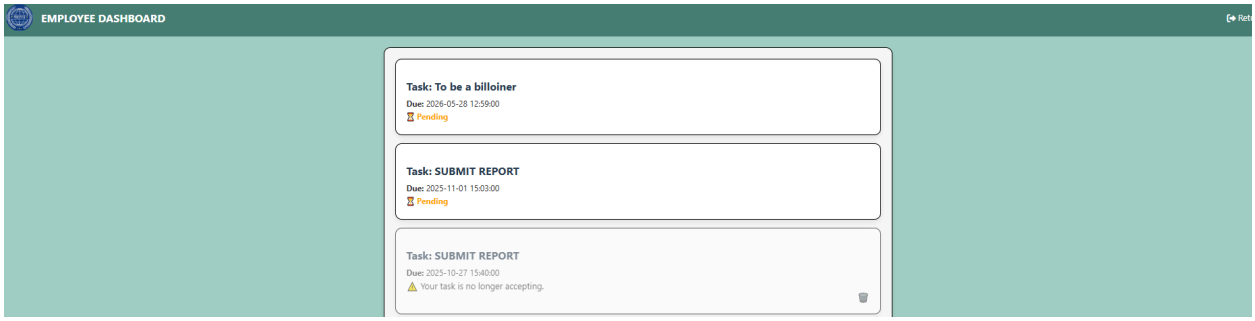
PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

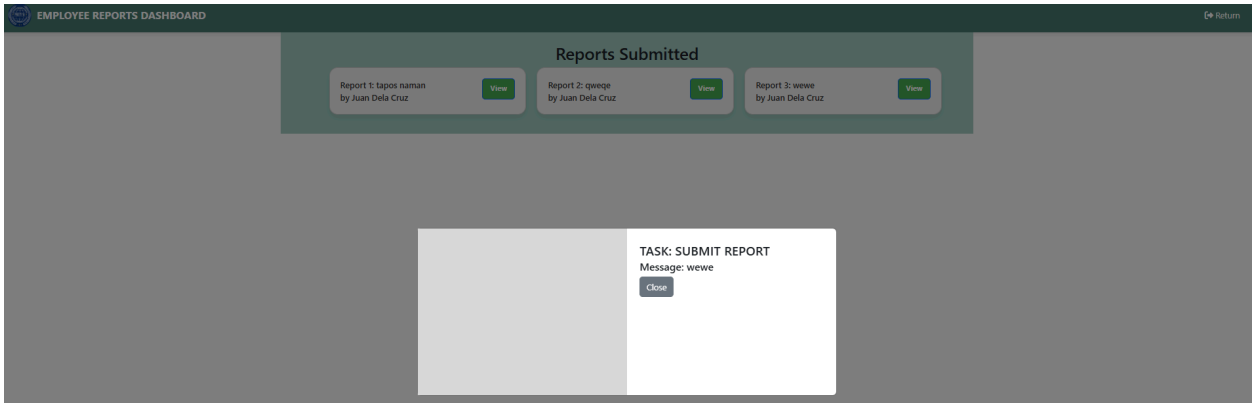
- **Attendance(Employees)**



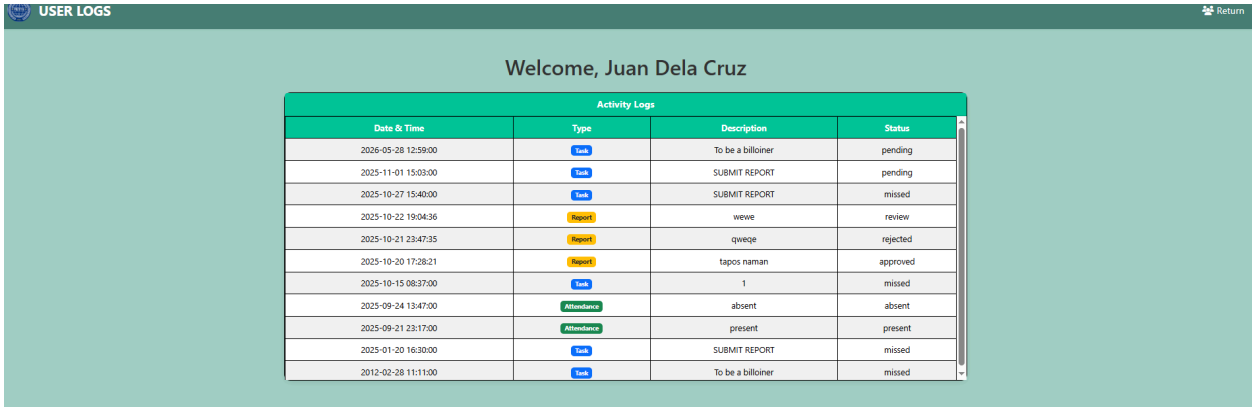
- **Assigned Task**



- **Created Report(Viewing)**

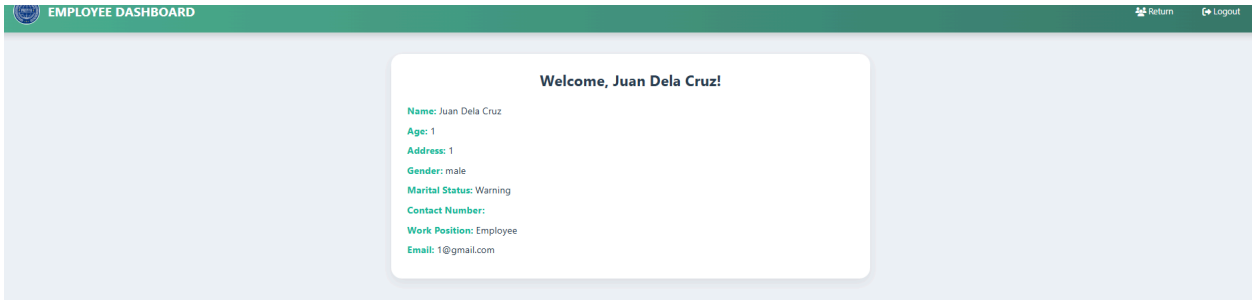


- **Activity Logs**

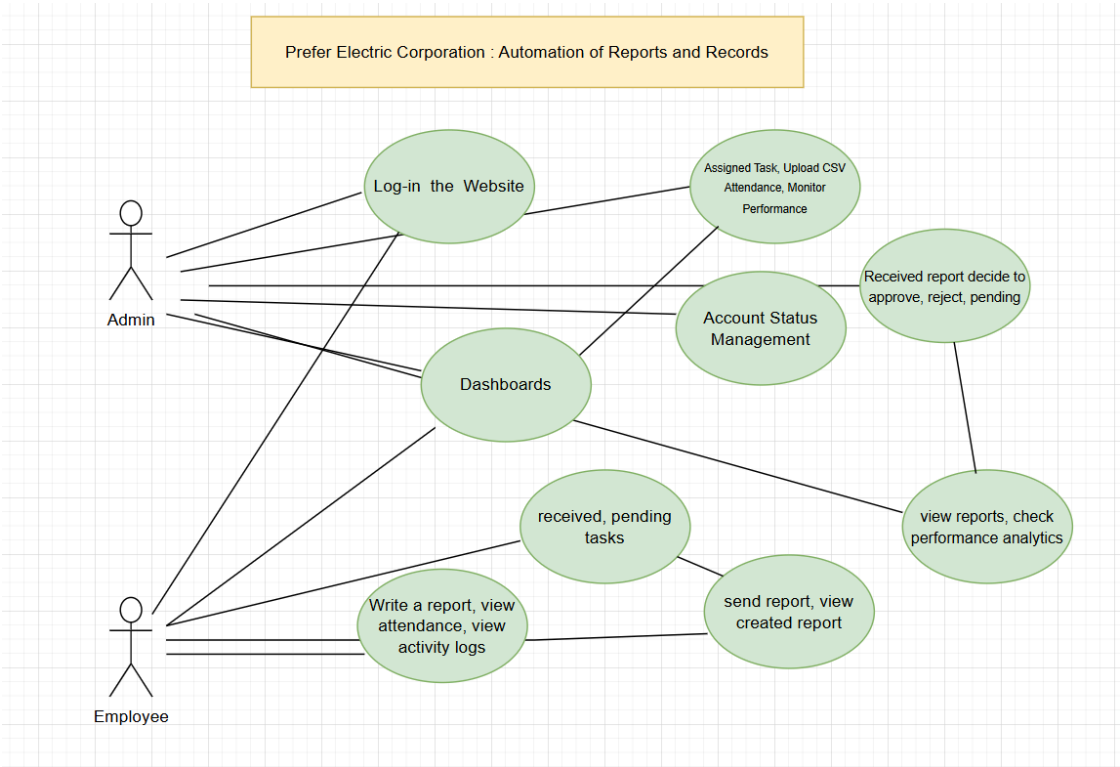


Date & Time	Type	Description	Status
2026-05-28 12:59:00	Task	To be a billoiner	pending
2025-11-01 15:03:00	Task	SUBMIT REPORT	pending
2025-10-27 15:40:00	Task	SUBMIT REPORT	missed
2025-10-22 19:04:36	Report	wewe	review
2025-10-21 23:47:35	Report	qweqe	rejected
2025-10-20 17:28:21	Report	tapos naman	approved
2025-10-15 08:37:00	Task	1	missed
2025-09-24 13:47:00	Attendance	absent	absent
2025-09-21 23:17:00	Attendance	present	present
2025-01-20 16:30:00	Task	SUBMIT REPORT	missed
2012-02-28 11:11:00	Task	To be a billoiner	missed

- **User's Information**



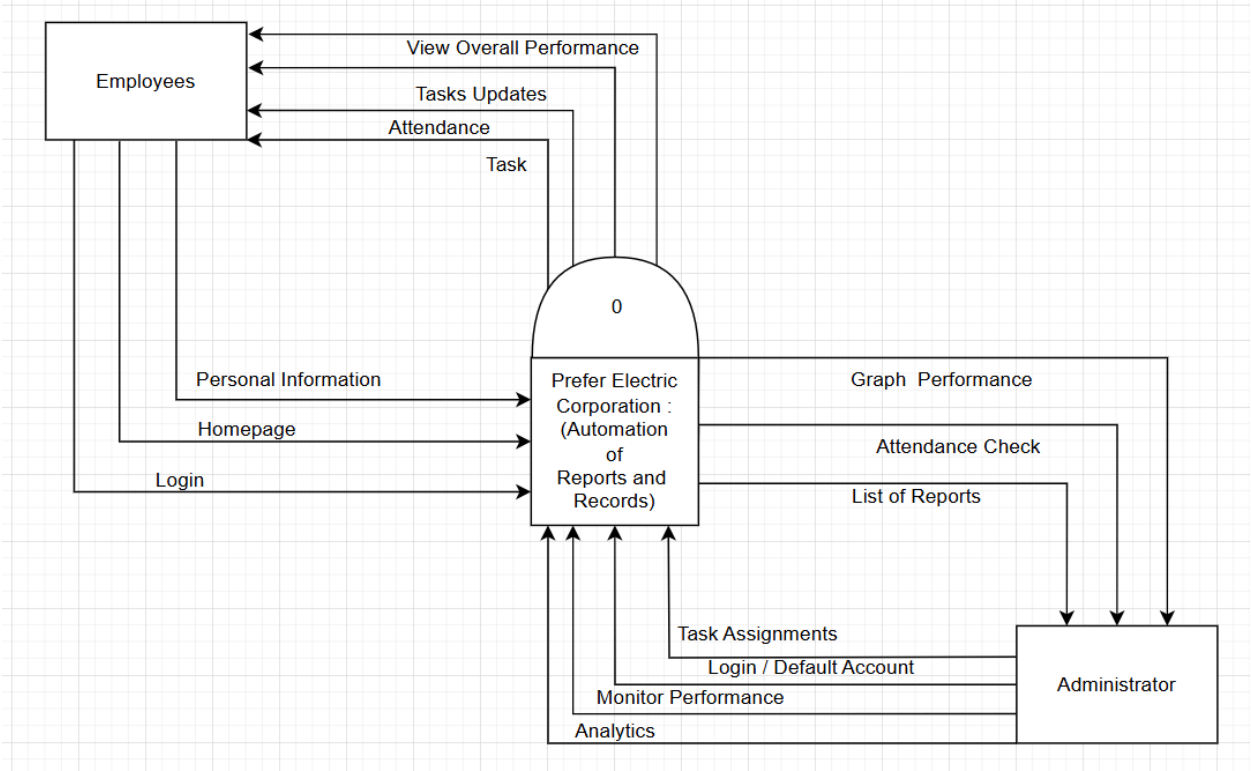
Appendix B :Use Case Diagram



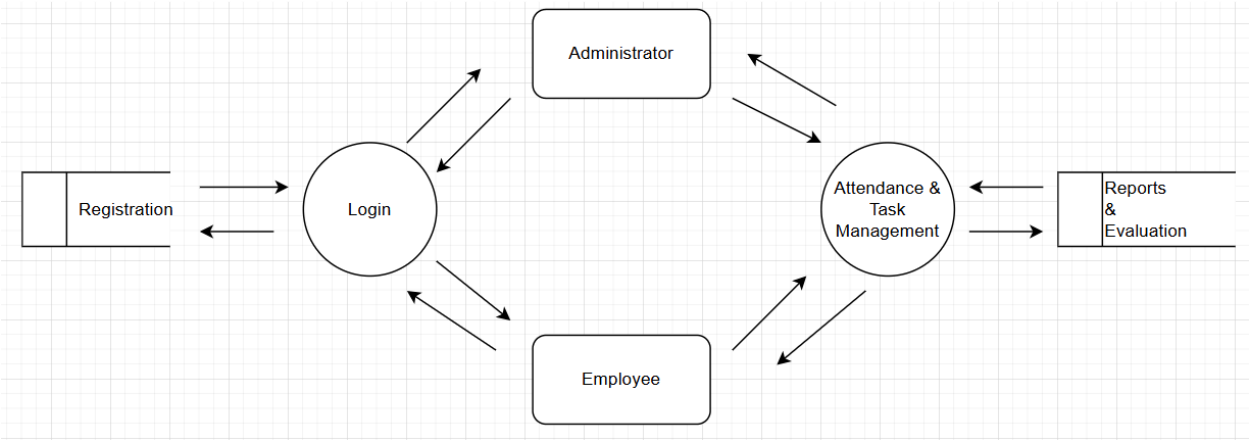
PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Appendix C :Data Flow Diagram



Data Flow Diagram Level 0



Data Flow Diagram Level 1

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Appendix D : User Survey Form and Summary

Demographic Information

Name: _____
Age: _____
Position/Department: _____

1.What is your role in the company?
- ☐ Employee/User
- ☐ Administrative Staff
- ☐ Department Head
- ☐ IT Personnel
- ☐ Other: _____

2.How often do you use the reporting or record management system?
- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Rarely

3.How familiar are you with the use of automated systems for reports and records management?
- ☐ Very familiar
- ☐ Somewhat familiar
- ☐ Heard about it but not familiar
- ☐ Not familiar at all

4.How important do you think it is to systematically organize and monitor company reports and records?
- ☐ Very important
- ☐ Important
- ☐ Neutral
- ☐ Slightly important

☐ Not important

5.Do you believe that using an automated system can improve report generation and record accuracy?

☐ Strongly agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly disagree

6.How would you rate the current process of handling reports and records in the company (manual or existing system)?

☐ Very efficient
☐ Efficient
☐ Neutral
☐ Inefficient
☐ Very inefficient

7.In your opinion, how necessary is it to replace the manual record-keeping process with an automated system?

☐ Very necessary
☐ Somewhat necessary
☐ Neutral
☐ Not necessary

8.Do you think an automated system can help reduce human error in managing reports and records?

☐ Strongly agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly disagree

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

9.How do you usually report issues or concerns related to reports or records?

- ☐ Inform immediate supervisor or admin staff
- ☐ Report to IT department
- ☐ Submit through a written form
- ☐ Do not report
- ☐ Other: _____

10.How long does it usually take before reported issues are addressed or resolved?

- ☐ Within the day
- ☐ 1-3 days
- ☐ More than 3 days
- ☐ Not sure

11.What challenges do you commonly experience in managing or accessing company records?

- ☐ Delayed reporting process
- ☐ Missing or misplaced files
- ☐ Data inconsistency or duplication
- ☐ Limited system access
- ☐ Lack of technical support
- ☐ Other: _____

12.How important is it to have security measures that protect company reports and confidential data?

- ☐ Very important
- ☐ Important
- ☐ Slightly important
- ☐ Not important

13.Do you believe that an automated system can improve accountability among employees and admin staff?

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral

- ☐ Disagree
- ☐ Strongly disagree

14. How confident would you feel if all company reports and records were stored and secured in a digital system?
- ☐ Very confident
- ☐ Confident
- ☐ Neutral
- ☐ Unconfident
- ☐ Very unconfident

15. How likely do you think the Automation of Reports and Records System will improve efficiency and organization in company operations?
- ☐ Very likely
- ☐ Likely
- ☐ Unlikely
- ☐ Very unlikely

16. How helpful would it be if the system could automatically generate reports for audits, employee performance, or daily summaries?
- ☐ Very helpful
- ☐ Helpful
- ☐ Slightly helpful
- ☐ Not helpful

17. Do you think proper training should be provided to all employees and staff on how to use the system?
- ☐ Strongly agree
- ☐ Agree
- ☐ Disagree
- ☐ Strongly disagree

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

18.How beneficial is it to have automated notifications or reminders for pending reports or record submissions?

- ☐ Very beneficial
- ☐ Beneficial
- ☐ Slightly beneficial
- ☐ Not beneficial

19.Do you believe that automation can promote transparency and accuracy in company documentation?

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

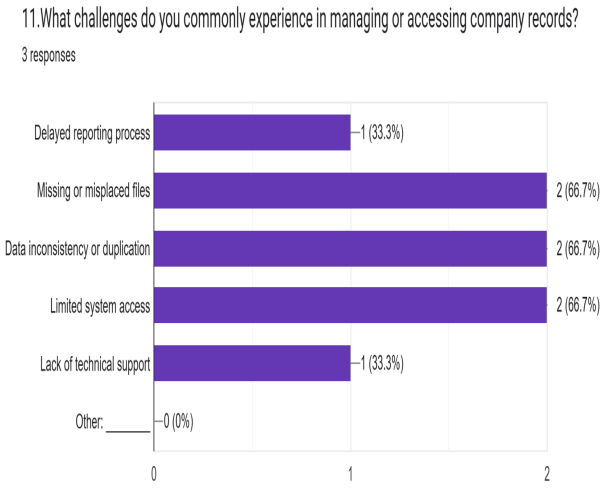
20.How important do you think implementing the Automation of Reports and Records System is for improving company operations?

- ☐ Very important
- ☐ Important
- ☐ Neutral
- ☐ Slightly not important
- ☐ Not important

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

- Summary of Response (e.g., tables, charts)



PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

John Lloyd Almonte
Malunang, Zarraga Iloilo
Iloilo ,5004
joar.almonte.ui@phinmaed.com
09512708341

October 30, 2025

Rafael Nazareta

AutoCAD Operator / Authorized Representative

Prefer Office Department
Prefer Electrical Engineering & Supply Corporation
Bigke, Leganes, Iloilo, 5003

Dear Mr. Nazareta,

Subject: Proposal for Memorandum of Agreement to Use the "Prefer Electric Corporation : Automation of Reports and Records"

We are excited to present our proposed Automation of Reports and Records System for Prefer Electric Corporation, which aims to enhance your organization's data management through automation, instant reports and secure digital records. This system is designed as a more effective, accurate and user-friendly option that replaces traditional and manual methodologies while ensuring departments have instant access to important information.

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

To formalize the implementation and use of this system, we propose an execution of a Memorandum of Agreement (MOA) that lays out roles, responsibilities and terms for both parties. This agreement will clearly define the scope, objectives, to secure data, and maintenance of the project, ensuring clarity and mutual understanding throughout the implementation and operation of the system.

The Automation of Reports and Records System will provide the following major benefits to Prefer Electric Corporation:

- Improved accuracy and reliability - decreased human error and inconsistencies in report writing.
- Streamlined reporting process - decrease data entry, report writing, and record changing through automation.
- Real time monitoring and access - allows authorized users to review and track information in real time.
- User-friendly - designed for employees to simplify tasks at every level.
- Performance and user evaluation functions - help you understand the system's efficiency and user satisfaction.

We are committed to providing full support throughout the implementation phase, which includes user training, technical support, and ongoing maintenance to ensure smooth addition to your workflow. Our goal is to make the Automation of Reports and

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Records System a dependable resource that enhances your record management process and contributes to the company's digital transformation.

We look forward to working with Prefer Electric Corporation to improve your operational efficiency and achieve increased accuracy and productivity with this new automation system. Please let us know the time you prefer for the conversation, or if you require any additional information.

Thank you for your consideration.

Sincerely,

John Lloyd Almonte
Project Manager / Web Developer
Prefer Electric Corporation ARR

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

PHINMA UNIVERSITY OF ILOILO
College of Information and Technology Education
Rizal Street, Iloilo City Proper, Iloilo City 5000, Philippines

MEMORANDUM OF AGREEMENT (MOA)

Between
Prefer Office Department of Prefer ELectrical Engineering & Supply Corporation
And
John Lloyd Almonte,

John Paul Anaud,

Date: October 30, 2025

Purpose

This agreement is entered into by and between the Prefer Office Department of Prefer Electrical Engineering & Supply Corporation, and the above-named students from PHINMA University of Iloilo, to collaborate on the students' capstone project entitled "Prefer Electric Corporation : Automation of Reports and Records".

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Scope of Agreement

The Prefer Office Department agrees to provide support and resources, including:

- Access to laptop,reports, and records
- Guidance and mentorship
- Relevant information for project development

The students agree to:

- Conduct research
- Develop and test the project
- Complete the capstone project according to requirements

Client Approval

By signing this document, the Prefer Office Department acknowledges approval and support of the students' proposed capstone project. The department commits to timely cooperation and to provide relevant information and access as needed for the project's success.

Signatures:

For the Prefer Office Department / Reports and Records :

_____

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Rafael Nazareta
AutoCAD Operator / Authorized Representative
Prefer Electrical Engineering & Supply Corporation
Date: _____

For the Students:

_____

John Lloyd Almonte
Capstone Project Student
Date: _____

_____

John Paul Anaud
Capstone Project Student
Date: _____

_____